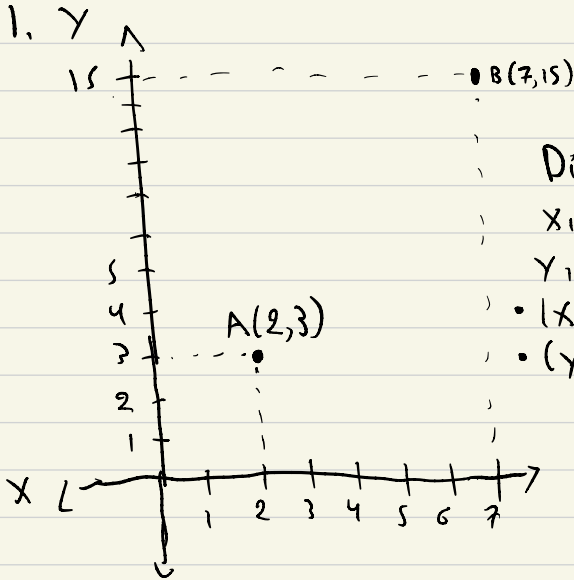


Problem Set 1

Section 1

Points $A(x_1, y_1)$ $B(x_2, y_2)$
 $A(2, 3)$ $B(7, 15)$



$$\text{Distance } AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$x_1 = 2 \quad x_2 = 7$$

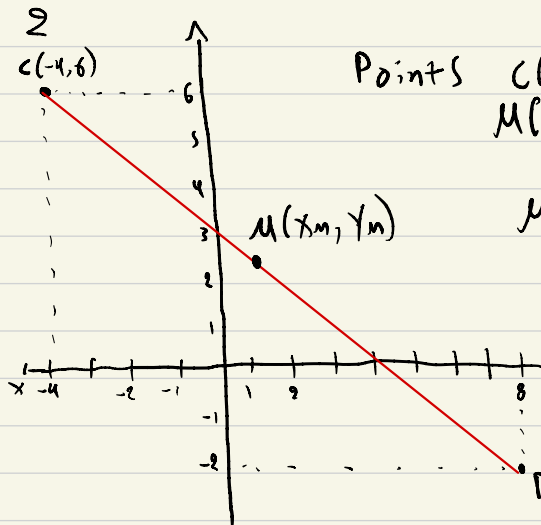
$$y_1 = 3 \quad y_2 = 15$$

$$\bullet (x_2 - x_1)^2 = (7 - 2)^2 = 5^2 = 25$$

$$\bullet (y_2 - y_1)^2 = (15 - 3)^2 = 12^2 = 144$$

$$\bullet AB = \sqrt{25 + 144} = \sqrt{169} = 13$$

So the distance between A, B is $AB = 13$



Points $C(-4, 6)$ $D(8, -2)$
 $M(x_m, y_m)$ midpoint of CD

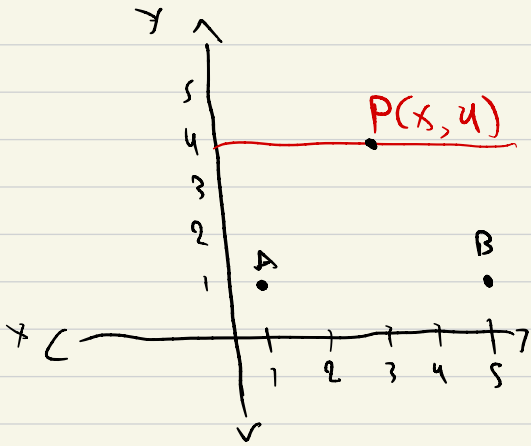
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \quad \begin{array}{ll} x_1 = -4 & x_2 = 8 \\ y_1 = 6 & y_2 = -2 \end{array}$$

$$M = \left(\frac{-4 + 8}{2}, \frac{6 + (-2)}{2} \right) = \left(\frac{4}{2}, \frac{4}{2} \right) =$$

$$M = (2, 2) = \text{midpoint of CD}$$

3. $P(x, 4)$ $A(1, 1)$ $B(5, 1)$

The distance $PA = PB$



$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$PA = \sqrt{(1 - x)^2 + (1 - 4)^2} = \sqrt{(1 - x)^2 + 9}$$

$$PB = \sqrt{(5 - x)^2 + (1 - 4)^2} = \sqrt{(5 - x)^2 + 9}$$

Because $PA = PB$ we have:

$$\sqrt{(1 - x)^2 + 9} = \sqrt{(5 - x)^2 + 9}$$

$$(1 - x)^2 + 9 = (5 - x)^2 + 9$$

$$(1 - x)^2 = (5 - x)^2$$

$$1 - 2x + x^2 = 25 - 10x + x^2$$

$$10x - 2x = 25 - 1$$

$$8x = 24$$

$$x = 24 / 8$$

$$x = 3$$

$$\text{So } P = (3, 4)$$

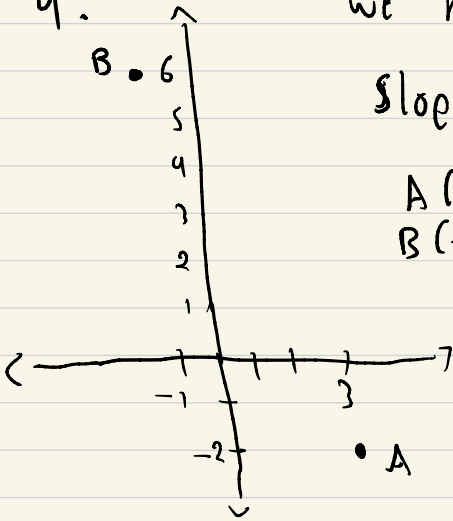
$$(a - b)^2 = a^2 - 2ab + b^2$$

• we separate x and numbers

• divide by 8

Section B

4.



we have $A(3, -2)$ $B(-1, 6)$

$$\text{slope } (m) = \frac{y_2 - y_1}{x_2 - x_1}$$

$$A(3, -2) \quad x_1 = 3 \quad y_1 = -2$$

$$B(-1, 6) \quad x_2 = -1 \quad y_2 = 6$$

$$\text{So } (m) = \frac{6 - (-2)}{-1 - 3}$$

$$(m) = \frac{6 + 2}{-4}$$

$$(m) = \frac{8}{-4}$$

$$(m) = -2 = \text{slope}$$

5.

$$A(2, 5)$$

$$\text{slope} = -3$$

we use $y = mx + c$

we know $m = -3$

we know $A(2, 5)$

$$\text{So } y = mx + c$$

$$5 = -3(2) + c$$

$$5 = -6 + c$$

$$11 = c$$

we replace c and m

$$\text{So } y = -3x + 11$$

And if we want to see if it's correct we put the $x = 2$ and $y = 5$

$$\bullet x = 2 \quad y = -3 \cdot 2 + 11 \Rightarrow y = -6 + 11 \Rightarrow y = 5$$

$$\bullet y = 5 \quad 5 = -3x + 11 \Rightarrow 5 - 11 = -3x \Rightarrow -6 = -3x \Rightarrow x = 2$$

6.



$$A(1, 2) \quad B(4, 11)$$

$$\bullet \text{ Slope } (m) = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{11 - 2}{4 - 1} = \frac{9}{3} = 3$$

$$\underline{m = 3}$$

$$\text{we use } y = mx + c$$

$$\text{we use } A(1, 2) \quad 2 = 3 \cdot 1 + c$$

$$\underline{-1 = c}$$

$$\text{so } \underline{y = 3x - 1} \quad (1)$$

To see if it's correct we use $B(4, 11)$ at (1)

$$\bullet x = 4 \quad y = 3 \cdot 4 - 1 \Rightarrow \underline{y = 11}$$

$$\bullet y = 11 \quad 11 = 3x - 1 \Rightarrow \underline{12 = 3x} \Rightarrow \underline{x = 4}$$

$$B(4, 11) \quad 11 = 3 \cdot 4 + c$$

$$11 = 12 + c$$

$$\underline{-1 = c}$$

It's the same c if we use B here

Section C

7. $A(50, 100)$ $B(200, 400)$ $t=5$

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$(x_2 - x_1)^2 = (200 - 50)^2 = 150^2 = 22500$$

$$(y_2 - y_1)^2 = (400 - 100)^2 = 300^2 = 90000$$

$$d = \sqrt{22500 + 90000} = \sqrt{112500} \approx 335.41$$

The character travels 335.41 pixels

$$\text{Speed (u)} = \frac{d}{t} = \frac{335.41}{5} \approx 67.08$$

The speed is 67.08 pixels/second

8. $E(300, 250)$ $t=20$
 $P(310, 265)$

we will find the distance
between E, P

$$\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (x_2 - x_1)^2 = (310 - 300)^2 = 100$$

$$(y_2 - y_1)^2 = (265 - 250)^2 = 225$$

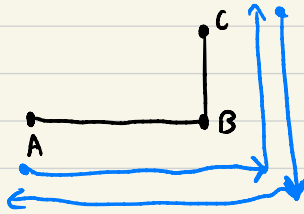
$$d = \sqrt{100 + 225} = \sqrt{325} \approx 18.03$$

Now if $d > r$ the player is out of the box
if $d < r$ the player is in the box

so $18.03 < 20$ the player is in the box

9. $A(100, 100)$ $B(100, 300)$ $C(400, 300)$

The path is



So we will find the distances of AB, BC

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d_{AB} = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$$

$$(x_B - x_A)^2 = (100 - 100)^2 = 0$$

$$(y_B - y_A)^2 = (300 - 100)^2 = 40000$$

$$\underline{d_{AB} = \sqrt{40000} = 200}$$

$$d_{BC} = \sqrt{(x_C - x_B)^2 + (y_C - y_B)^2}$$

$$(x_C - x_B)^2 = (400 - 100)^2 = 90000$$

$$(y_C - y_B)^2 = (300 - 300)^2 = 0$$

$$\underline{d_{BC} = \sqrt{90000} = 300}$$

The total distance will be the sum
of $(d_{AB} + d_{BC}) \cdot 2$ because the NPC does
the same distance 2 times, one time from A to C
and back again from C to A

$$\text{So total } d = (200 + 300) \cdot 2 = 500 \cdot 2 = \underline{1000}$$

10. $P_1(120, 80)$ $P_2(480, 240)$

$$M_{\text{midpoint}}(M) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{so } M_{P_1, P_2} = \left(\frac{120 + 480}{2}, \frac{80 + 240}{2} \right)$$

$$= \left(\frac{600}{2}, \frac{320}{2} \right) = (300, 160)$$

$$\underline{M = (300, 160)}$$

11.

$$T(0, 0) \quad E(6, 8)$$

a) $\text{Slope } (m) = \frac{y_2 - y_1}{x_2 - x_1}$

$$m_{TE} = \frac{8 - 0}{6 - 0} = \frac{8}{6} = \frac{4}{3} \approx 1.33 = m_{TE}$$

b) $\text{distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$d_{TE} = \sqrt{(x_E - x_T)^2 + (y_E - y_T)^2}$$

$$(x_E - x_T)^2 = (6 - 0)^2 = 36$$

$$(y_E - y_T)^2 = (8 - 0)^2 = 64$$

$$d_{TE} = \sqrt{100} = 10 = d$$